

Degree vs. Skill: Are FJWU Students Truly Job-Ready?

A Data Analytics Perspective on Academic Performance and Employability

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Abstract

The growing disconnect between academic achievement and professional preparedness has become a critical concern in higher education, particularly within technology disciplines. This study investigates whether Computer Science students at Fatima Jinnah Women University (FJWU) possess the practical competencies required by the modern software industry, beyond the conventional measure of cumulative grade point average (CGPA).

A survey-based dataset of 108 student responses was analyzed using data analytics, inferential statistics, and Natural Language Processing (NLP) techniques. The findings indicate that academic performance is a weak predictor of self-reported job readiness. Students with internship experience, external certifications, GitHub portfolios, and project-based learning backgrounds demonstrated substantially higher confidence and practical competence than peers who relied solely on formal coursework.

The study further identifies significant deficiencies in communication skills, DevOps practices, project management, and cloud technologies, competencies that are increasingly prioritized by industry recruiters. The results are presented through statistical analysis, data visualizations, NLP-derived sentiment insights, and an interactive dashboard interface designed for both technical and non-technical audiences.

1. Introduction

The global technology industry is undergoing rapid transformation, driven by advances in cloud computing, DevOps culture, agile software development, and artificial intelligence. As a result, employer expectations for entry-level software professionals have expanded well beyond academic credentials. Organizations increasingly evaluate candidates on practical experience, portfolio work, communication ability, and adaptability qualities that are not always reflected in a student's academic transcript.

In Pakistan's technology sector, this gap between graduate output and industry demand is particularly pronounced. Universities produce a large number of Computer Science graduates each year, yet many struggle to secure relevant employment despite strong academic records. This creates an important question worth examining: does a strong CGPA actually prepare students for the workforce, or does it serve primarily as an institutional benchmark disconnected from real-world competency?

This project was designed to address that question in a specific, grounded context — the Computer Science students of Fatima Jinnah Women University, Rawalpindi. Through a structured survey, statistical modeling, and NLP-based analysis of student feedback, this research attempts to quantify the gap between academic performance and industry readiness, and to identify the factors that most meaningfully contribute to a student's employability.

2. Problem Statement

A consistent pattern has emerged in graduate employment outcomes: students with high academic scores frequently report feeling underprepared for professional roles, while students with moderate GPAs but active portfolios and practical experience demonstrate stronger performance in recruitment settings. This suggests that the current model of academic evaluation may inadequately represent professional capability.

The central question motivating this research is:

Does academic performance alone determine job readiness, or do practical skills and industry exposure play a more significant role in a student's employability?

To investigate this, the study examines the relationship between CGPA and practical skill proficiency, the impact of internship participation on student confidence, the role of certifications in bridging curriculum gaps, and the specific technical and soft skills most commonly underdeveloped among graduating students. Additionally, student sentiments and career concerns expressed through open-ended survey responses are analyzed to provide a qualitative dimension to the data.

3. Research Objectives

This study is guided by the following objectives:

1. To examine whether a statistically significant relationship exists between CGPA and self-rated job readiness among FJWU Computer Science students.
2. To assess the level of industry exposure through internships, freelancing, open-source contribution, and portfolio activity — within the student population.
3. To identify specific technical and interpersonal skill gaps relative to current industry expectations.
4. To apply Natural Language Processing techniques to open-ended survey responses, extracting recurring themes, sentiments, and concerns related to career preparedness.
5. To develop an interactive, accessible analytics dashboard that communicates findings to both student and institutional audiences.
6. To offer evidence-based recommendations for students, academic departments, and university administration.

4. Scope and Limitations

This study focuses specifically on undergraduate Computer Science students at Fatima Jinnah Women University, Rawalpindi, and draws on a dataset of 108 survey responses collected during the 2025 academic year. The analysis covers academic performance metrics, technical skill self-assessments, professional exposure indicators, and open-ended career reflections.

The scope does not extend to students at other institutions, and no external recruiter or employer data was incorporated. As the data is self-reported, findings are subject to inherent subjectivity and social desirability bias. The sample size, while sufficient for exploratory

analysis, limits the generalizability of results to broader populations. These constraints are acknowledged throughout the analysis, and future work is recommended to address them through expanded datasets and longitudinal tracking.

5. Literature Review

Research in higher education and workforce development has consistently highlighted a misalignment between university curricula and industry expectations. Studies from diverse academic contexts suggest that graduates with applied experience through internships, collaborative projects, or independent portfolio work tend to achieve stronger employment outcomes than those with comparable academic records but limited practical exposure [citation].

Within the software engineering discipline, this gap is especially well-documented. Industry surveys conducted by major technology employers consistently rank skills such as version control proficiency, collaborative development practices, cloud infrastructure familiarity, and communication ability among the most valued competencies in new hires none of which are reliably captured by GPA alone [citation].

In the South Asian higher education context, the issue is compounded by curricula that often emphasize theoretical instruction over applied practice. Students may graduate with strong knowledge of algorithms and formal computing theory while having limited exposure to modern development toolchains, agile workflows, or real-world software engineering environments [citation].

Pakistan's technology sector, while growing rapidly, continues to face a talent gap that is attributed in part to this academic-industry disconnect. Employers in cities such as Karachi, Lahore, and Islamabad frequently report difficulty finding graduates who are technically current and professionally prepared, despite the large volume of annual CS graduates [citation].

This project contributes to the existing body of literature by providing a localized, data-driven investigation into these dynamics within a women's university in Rawalpindi, offering institutional-level insights that are underrepresented in existing research.